

Waste Material Classification using MobileNetV2

Technical Analysis Document

Introduction to Computer Vision - Final Project

Task: Multi-class image classification

Architecture: MobileNetV2 with transfer learning

Classes: Cardboard, Glass, Metal, Paper, Plastic

Academic technical report covering problem statement, methodology, experimental results, failure analysis, interpretability, and ethical considerations.

1. Problem Statement

Waste segregation is an important environmental task in which discarded objects must be assigned to the correct material category before recycling or disposal. Manual sorting can be slow and inconsistent, while visual appearance varies because of object shape, illumination, background, deformation and contamination. The objective of this project is to develop a reproducible Computer Vision system that automatically classifies photographs of waste into five common material categories.

The system receives an RGB image and predicts one of five labels: cardboard, glass, metal, paper or plastic. The project covers the complete workflow from dataset preparation to quantitative evaluation and visual interpretation.

1.1 Objectives

- Prepare a deterministic training, validation and test split.
- Use learned CNN features through a pretrained MobileNetV2 backbone.
- Train a custom five-class classification head using transfer learning.
- Evaluate the system with Accuracy, Precision, Recall, F1-score and a Confusion Matrix.
- Provide single-image inference with confidence scores and complete class probabilities.
- Use Grad-CAM to inspect the image regions influencing predictions.

1.2 Dataset

The project uses the Garbage Dataset - A Comprehensive Image Dataset for Garbage Classification and Recycling. The supplied dataset description reports 13,348 images across 10 classes: Metal (930), Glass (1,736), Biological (699), Paper (1,336), Battery (756), Trash (453), Cardboard (1,411), Shoes (1,449), Clothes (1,892), and Plastic (1,597). The dataset is distributed under the MIT License.

Five material-oriented classes were selected for this project: Cardboard, Glass, Metal, Paper and Plastic. The preprocessing script creates a 70% training, 15% validation and 15% test split using random seed 42. The resulting held-out test set contains 1,056 images.

2. Methodology

2.1 Preprocessing and Data Augmentation

Images are resized to 224 x 224 pixels. Training images use random horizontal flipping, random rotation and ColorJitter to increase visual diversity. Validation and test images are resized without random augmentation. All images are converted to tensors and normalized using ImageNet mean and standard deviation values.

2.2 Feature Representation and Model

The project uses learned convolutional features rather than handcrafted descriptors. MobileNetV2 pretrained on ImageNet is used as the feature extractor. The convolutional backbone is frozen in the implemented transfer-learning stage and the original classifier is replaced with a custom head containing Dropout with probability 0.3 followed by a Linear layer mapping 1,280 features to five output classes.

2.3 Training Configuration

Parameter	Value
Architecture	MobileNetV2
Pretraining	ImageNet
Classes	5
Image size	224 x 224
Batch size	32
Optimizer	Adam
Learning rate	0.001
Loss	CrossEntropyLoss
Classifier	Dropout(0.3) + Linear(1280, 5)
Random seed	42

MobileNetV2 was selected because it provides a strong compromise between classification performance and computational efficiency, making it suitable for transfer learning and CPU-based experimentation.

2.4 Post-processing

During inference, network logits are converted to probabilities using softmax. The class with the highest probability is selected as the prediction. The prediction module reports the predicted class, confidence score and full probability distribution. Grad-CAM is then available as an interpretability stage to visualize influential image regions.

3. Implementation and Reproducibility

The implementation is modular and divided into five Python scripts. `prepare_data.py` creates the dataset split and dataset summaries. `train.py` defines transformations, DataLoaders, MobileNetV2, loss function and optimizer, performs training and validation, and stores the best checkpoint. `evaluate.py` evaluates the held-out test set. `predict.py` performs single-image inference. `gradcam.py` generates visual explanations.

The repository includes `requirements.txt` for dependency installation, `README.md` for project documentation, `models/` for trained checkpoints and `results/` for generated metrics and visual outputs. Large local datasets, generated results, trained models and the virtual environment can be excluded from version control through `.gitignore`.

3.1 Reproducibility Test

A clean Python virtual environment was recreated and the project dependencies were installed successfully from `requirements.txt`. The complete pipeline was then validated. Dataset preparation, model evaluation, single-image prediction and Grad-CAM all completed successfully.

3.2 Validated Inference Example

For the test image `plastic_1.jpg`, the system predicted plastic with 67.43% confidence. The remaining probabilities were glass 26.14%, cardboard 2.27%, metal 2.14% and paper 2.02%. This demonstrates that the inference stage reports both the final label and useful uncertainty information.

4. Experimental Results

The final trained model was evaluated on 1,056 unseen test images. The evaluation computes Accuracy, Precision, Recall, F1-score, a complete classification report and a confusion matrix.

Metric	Result
Accuracy	82.39%
Macro Precision	81.94%
Macro Recall	82.68%
Macro F1-score	81.97%
Misclassified images	186 / 1056

An accuracy of 82.39% means that the model correctly classifies more than four out of five unseen test images. Macro precision, recall and F1-score remain close to the overall accuracy, indicating reasonably balanced performance across the five categories.

4.1 Per-Class Performance

Class	Precision	Recall	F1-score	Support
Cardboard	0.8894	0.8310	0.8592	213
Glass	0.8450	0.8774	0.8609	261
Metal	0.6994	0.8643	0.7732	140
Paper	0.7877	0.8308	0.8087	201
Plastic	0.8756	0.7303	0.7964	241

Glass has the highest recall (87.74%), while cardboard has the highest precision (88.94%). Metal shows relatively low precision (69.94%) despite high recall (86.43%). Plastic has high precision (87.56%) but the lowest recall (73.03%).

4.2 Confusion Matrix Analysis

True / Pred.	Cardboard	Glass	Metal	Paper	Plastic
Cardboard	177	1	1	30	4
Glass	0	229	16	2	14
Metal	6	8	121	3	2
Paper	13	6	10	167	5
Plastic	3	27	25	10	176

The largest single confusion is cardboard predicted as paper, with 30 cases. Plastic is frequently confused with glass (27 cases) and metal (25 cases). Glass is sometimes classified as metal (16) or plastic (14). These patterns are consistent with similarities in texture, transparency, reflections and object appearance.

5. Failure Analysis

A total of 186 test images were misclassified. Cardboard and paper can share fibrous textures, flat geometry and printed surfaces. Transparent or reflective plastic may resemble glass, while reflective packaging can resemble metal. The classifier may also exploit contextual cues such as background and illumination rather than material composition alone.

The evaluation script saves `misclassified_images.csv` and `test_predictions.csv`, enabling systematic inspection of individual failure cases and supporting future error-driven improvements.

6. Interpretability with Grad-CAM

Grad-CAM visualizes the spatial regions contributing to a prediction. The implementation targets `model.features[-1]`, the final MobileNetV2 feature stage. The generated activation map is resized to 224 x 224 and normalized between 0 and 1.

For each analysed image, the script saves the original image, Grad-CAM heatmap, heatmap overlay and combined visualization. This provides a qualitative check of whether the network focuses on the waste object itself or on irrelevant image regions. Grad-CAM improves transparency but should not be interpreted as proof that the network has learned causal material properties.

7. Limitations and Future Improvements

- Only five of the ten available dataset categories are used.
- The classes are not perfectly balanced.
- The implemented transfer-learning configuration keeps the MobileNetV2 feature extractor frozen.
- Image-level classification assumes one dominant material per image.
- Performance on an independent external dataset has not been measured.

Future work could include fine-tuning later MobileNetV2 blocks, class weighting, training on all ten categories, architecture comparisons with ResNet or EfficientNet, independent external testing, real-time inference and mobile deployment.

8. Ethical, Privacy and Security Considerations

The project processes waste-object images rather than identity, medical or financial information, so direct privacy risk is limited. Nevertheless, dataset provenance and licensing must be respected, and incidental background information in photographs should be considered when datasets are collected or redistributed.

Dataset bias remains important. Lighting, camera type, geographical context, packaging design and waste condition may differ from the training distribution. Unequal class representation can also affect model behaviour.

The reported 82.39% test accuracy is appropriate for an academic prototype but does not justify autonomous industrial deployment. Incorrect predictions could contaminate recycling streams. A real-world system should include external validation, confidence thresholds, monitoring and potentially human review for uncertain predictions.

9. Conclusion

This project implements a complete and reproducible Computer Vision pipeline for five-class waste material classification. It combines deterministic dataset preparation, augmentation and normalization, MobileNetV2 transfer learning, quantitative evaluation, single-image prediction and Grad-CAM interpretability.

The final model achieves 82.39% accuracy, 81.94% macro precision, 82.68% macro recall and 81.97% macro F1-score on 1,056 held-out images. Failure analysis identifies meaningful confusion patterns between visually similar materials and provides clear directions for future improvement.

10. References

- Garbage Dataset - A Comprehensive Image Dataset for Garbage Classification and Recycling. Dataset description supplied for the project; 13,348 images, 10 classes, MIT License.
- The Garbage Dataset (GD): A Multi-Class Image Benchmark for Automated Waste Segregation.
- Managing Household Waste Through Transfer Learning.
- Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., Chen, L.-C. MobileNetV2: Inverted Residuals and Linear Bottlenecks. CVPR, 2018.
- Selvaraju, R. R. et al. Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization. ICCV, 2017.